The AP Research Poster Project

from Summer Training Materials
with student samples

Overview

- Create a poster that visually demonstrates the status of components of a Research project.
- Use the posters in short presentations about the progress of the work.
- Provide actionable feedback to peers.

STAGE 1
Exploring a Topic

Draft Research Title

Problem Statement/Summary of the topic being examined

Draft of Research Question

Potential Main Sources

Significance and/or Value of the Study

Places to Look for Help
(potential advisers, community organizations, libraries etc)

STAGE 2

Drafting an Inquiry Proposal

Draft Proposal Title

Problem Statement/Summary of the topic being examined

Draft of Research Question

Definitions and Assumptions

Data Sources and Places to Look for Help

Significance and/or Value of the Study

Proposed Method of Inquiry

STAGE 3
Finalizing an Inquiry Proposal

Proposal Title

Problem Statement/Summary of the topic being examined

Research Question

Definitions

Assumptions and Hypotheses

Significance and/or Value of the Study

Method of Inquiry and Example Study

Situatedness

Data Sources

Next Steps

What am I Missing?

Potential Issues

STAGE 4

Launching into Data Collection

Proposal Title

Problem Statement/Summary of the topic being examined

Research Question

Definitions

Assumptions and Hypotheses

Significance and/or Value of the Study

Method of Inquiry

Rationale for Method

Situatedness

Data Sources (Lit Review)

Data Sources (Evidence)

Ethical Considerations

Next Steps

What am I Missing?

Potential Issues

Stage 1

9/6/2016

Draft Research Title

The Manufacturing of Kpop Songs: Formula and Success of Music

Problem Statement

There are a multitude of theories concerning the supposed artificiality and similarity of various Kpop songs. It is uncertain if these songs fit into different assigned formats, and if their resemblance to the formula has any relation to success.

Draft Research Question

Can a common formula be determined for certain Kpop songs, and if so, does a song's resemblance to that defined formula have any correlation with the popularity or success of the song?

Potential Sources

The Social Mediascape of Transnational Korean Pop Culture.

Manufacturing Creativity: Production, Performance, and Dissemination of Kpop

Mass Media Technologies and Popular Music Genres

Significance / Value of Study

As the popularity of Kpop has expanded in recent years, there have been comments about it seeming artificial and manufactured. This study seeks to see if there is a common formula utilized by these songs, and if songs resembling the formula have higher levels of success. This has implications for finding the success of pop songs in general.

Places to look for help

YouTube, Social Media, Websites of Companies

JYP, SMTOWN, 1theK

Stage 2

AP Research [redacted] Draft Proposal Title

Manufacturing K-pop: The Formula of Music

Problem Statement

This study aims to examine different popular K-pop songs, and the variety of formats and types that they fit into. It also seeks to find a common pattern for popular songs, especially in terms of tempo, key, energy, etc.

Research Question

What different formats or formulas can be determined and generalized for popular K-pop songs, in terms of structure and type, and what patterns can be discovered when their attributes are analyzed?

Definitions / Assumptions

There is an existing pattern found within popular Kpop songs.

This pattern can be determined from Spotify Web API analysis

K-pop: Korean pop music

Data Sources

Spotify Web API (Echo Nest)

YouTube Music Videos (JYP, 1thek, SM Town, YG, Starship, TS, kt)

Company Websites

Significance

In recent years, the Korean pop music industry has expanded significantly, growing rapidly in terms of international success. The study aims to find repeating characteristics of the most successful and popular songs, which could prove invaluable for predicting a song's success and potential.

Proposed Method of Inquiry

Analysis of the most popular (in terms of MV views, sales, music show wins, etc.) K-pop songs over the last 5 years via the Spotify Web API.

Stage 3

Title: Manufacturing Kpop: The Formula for Music

Problem Statement

This study aims to examine different Kpop songs, and the different formats and types that they seem to fit into. It also seeks to quantify common patterns that different popular songs fit into, especially in terms of rhythm, beat, length, etc

Research Question

What different formats can be determined for popular and successful Kpop songs, in terms of structure and type, and what common patterns in the songs' musical qualities can be drawn?

Definitions

Kpop: Korean pop music

Popularity: Scaling variable based on different factors (MV views, show wins, album sales, etc.)

Assumptions / Hypothesis

Assumption: Kpop songs can be categorized into particular formats with differences to form a framework.

Hypothesis: Such a framework can be defined.

Significance / Value

In recent years, the Kpop music industry has expanded significantly, growing rapidly in terms of domestic and especially international success. This study aims to analyze the music produced in Korea, and to see the defining characteristics of the songs. This holds valuable information for song producers in Korea, and how they can improve their songs' successes.

Method of Inquiry and Example Study

correlational variable analysis of the most popular Kpop songs over the last few years through the Spotify Web API. This will be performed with measurements of the song and different scales of popularity.

Example Study: Visualizing a Hit, Shaun Ellis and Tom Engelhardt

Situatedness

Music, as a topic for study, has always been fairly difficult to measure and research in a quantitative method. However, with new technology, researchers have been able to analyze music, and I plan on applying this type of analysis to Korean pop music.

Data Sources

YouTube (music video views)

Music Shows (KBS, Mnet, Inkigayo - number of wins)

Spotify Web API (Echo Nest)

Gaon Album Chart (album sales, physical and digital)

Next Steps

Finalizing a quantitative method to scale the various measures and factors of a song's financial

What am I missing?

Missing a definitive way to determine which songs should be analyzed and included with the

Potential Issues

Issues could arise when trying to rationalize the choices made during research, due to there being limited available literature to establish a precedent.

Stage 4

THE K-POP FORMULA

PROBLEM STATEMENT

THIS STUDY AIMS TO EXAMINE DIFFERENT K-POP SONGS, AND THE DIFFERENT FORMATS THEY FIT INTO. IT ALSO SEEKS TO QUANTIFY COMMON PATTERNS THAT POPULAR SONGS SHARE, ESPECIALLY IN TERMS OF RHYTHM, BEAT AND LENGTH.

RESEARCH QUESTION

WHAT DIFFERENT FORMATS CAN BE DETERMINED FOR POPULAR K-POP SONGS, IN TERMS OF STRUCTURE, AND WHAT COMMON PATTERNS IN THE SONGS' MUSICAL QUALITIES CAN BE FOUND?

DEFINITIONS

K-POP: KOREAN POP MUSIC

POPULARITY: MEASURED WITH DIFFERENT FACTORS (MV VIEWS, MUSIC SHOW WINS, SALES, ETC.)

ASSUMPTIONS/HYPOTHESIS

ASSUMPTION: K-POP SONGS CAN BE CATEGORIZED INTO PARTICULAR FORMATS, AND DISTINCT DIFFERENCES CAN BE FOUND TO DEVELOP A FRAMEWORK.

HYPOTHESIS: THE MOST SUCCESSFUL SONGS IN EACH FORMAT WILL GRAVITATE TOWARDS SIMILAR QUALITIES, PARTICULARLY TEMPO AND ENERGY.

METHOD OF INQUIRY

MY METHOD IS A MIXED METHOD, CORRELATIONAL VARIABLE ANALYSIS OF THE MOST POPULAR K-POP SONGS OVER THE LAST 4 YEARS (2013 – 2016) THROUGH THE SPOTIFY WEB API, PERFORMED WITH MEASUREMENTS OF THE SONGS AND DIFFERENT SCALES OF POPULARITY. SONGS WITH AT LEAST ONE MUSIC SHOW WIN ARE SELECTED FOR MY ANALYSIS.

EXAMPLE STUDY: VISUALIZING A HIT, SHAUN ELLIS AND TOM ENGELHARDT (RUTGERS)

SIGNIFICANCE

IN RECENT YEARS, THE KOREAN POP MUSIC INDUSTRY HAS EXPANDED SIGNIFICANTLY, GROWING RAPIDLY IN TERMS OF FINANCIAL SUCCESS. THIS STUDY AIMS TO ANALYZE THE MUSIC PRODUCED BY THE VARIOUS ARTISTS IN KOREA, AND TO SEE IF THERE ARE ANY DEFINING CHARACTERISTICS OF POPULAR SONGS, WHICH IS VALUABLE INFORMATION FOR THE INDUSTRY AS A WHOLE. ADDITIONALLY, THIS METHOD OF ANALYZING MUSIC CAN BE GENERALIZED TO OTHER GENRES OF MUSIC AS WELL.

SITUATEDNESS

MUSIC, AS A TOPIC FOR STUDY, IS FAIRLY DIFFICULT TO MEASURE AND RESEARCH IN A QUANTITATIVE METHOD AND THE LIMITED NUMBER OF AVAILABLE LITERATURE REFLECTS THAT DIFFICULTY. HOWEVER, WITH MODERN TECHNOLOGY, IN PARTICULAR THE SPOTIFY WEB API, MUSIC CAN BE QUANTIFIED, AND I CAN APPLY THIS PROCESS TO STUDY THE KOREAN MUSIC INDUSTRY.

NEXT STEPS

THE NEXT STEP I NEED TO TAKE INVOLVES DETERMINING THE QUALITATIVE FORMATS I PLAN ON DIVIDING THE SONGS INTO.


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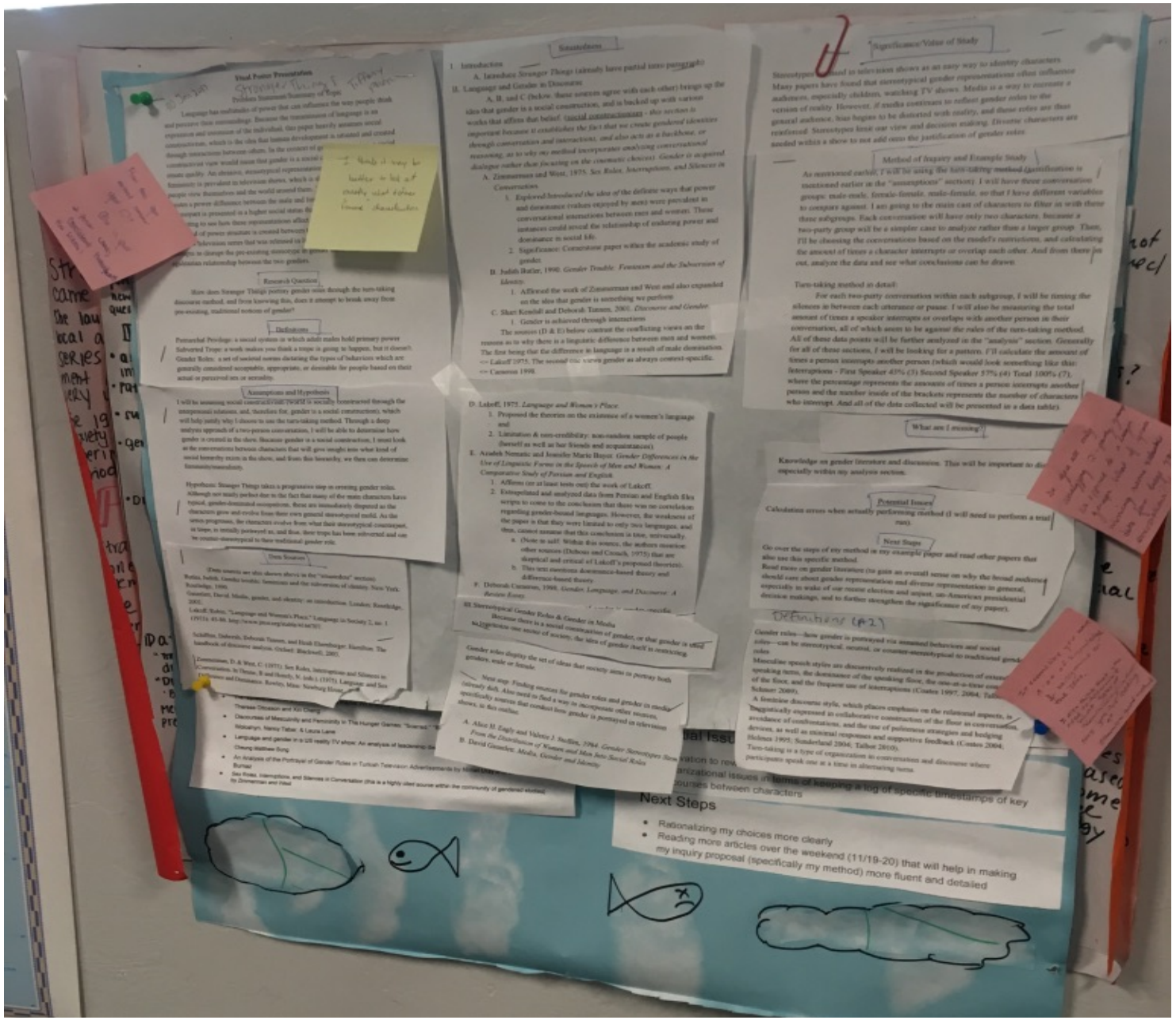
DATA SOURCES

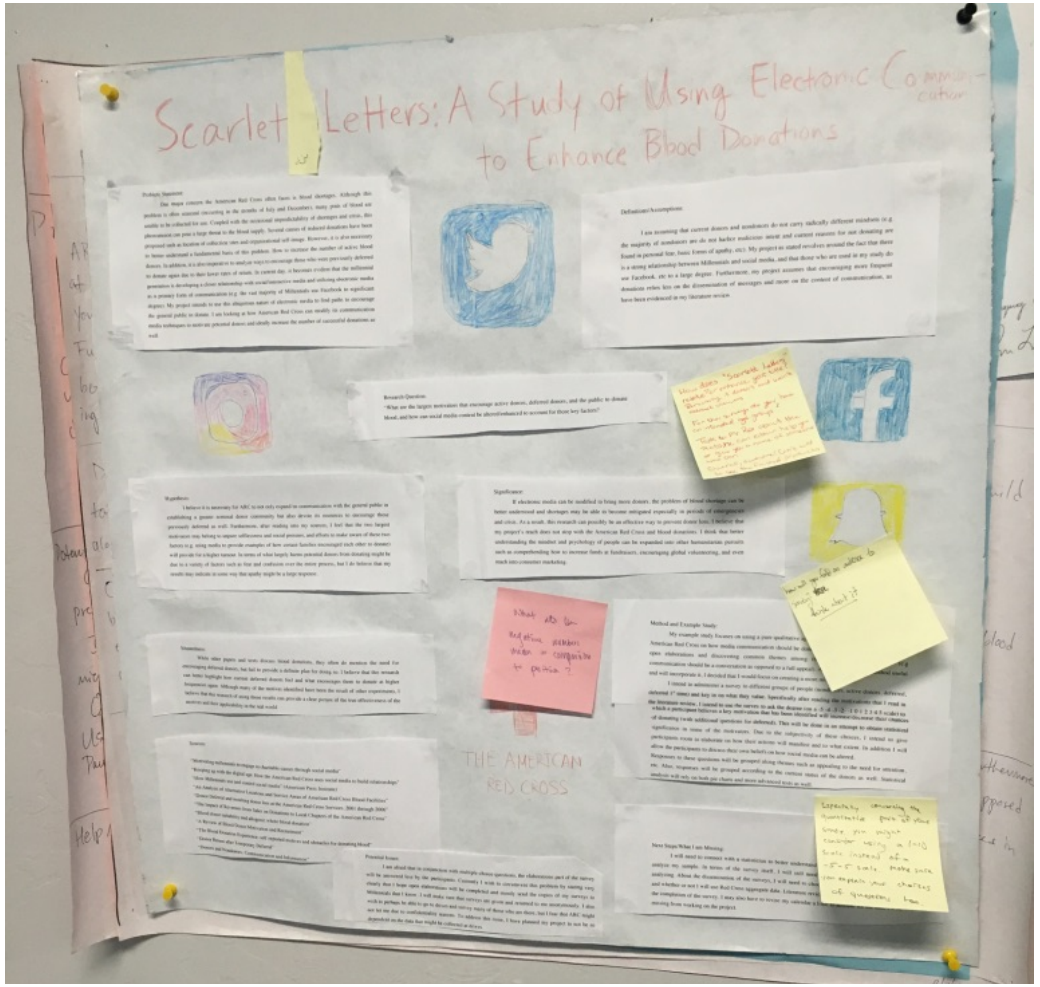
YOUTUBE (MUSIC VIDEO VIEWS)

MUSIC SHOWS (KBS, MNET, INKIGAYO – NUMBER OF WINS)

SPOTIFY WEB API (ECHO NEST)

GAON ALBUM CHART (ALBUM SALES, PHYSICAL AND DIGITAL)





Other Stage 4 examples with gallery walk feedback